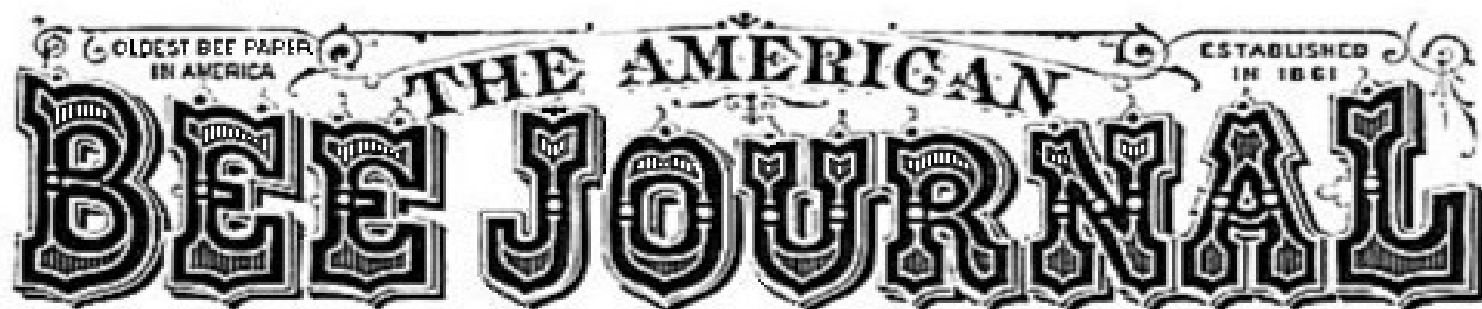




DEVOTED TO SCIENTIFIC BEE-CULTURE AND SALE OF PURE HONEY.

VOL. XVII.

CHICAGO, ILL., MARCH 23, 1881.

No. 12.

THE AMERICAN BEE JOURNAL

OLDEST BEE PAPER
IN AMERICA

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For the American Bee Journal

CORRESPONDENCE.

For the American Bee Journal

What is the Royal Jelly?

C. J. ROBINSON.

I propose, by permission, to discuss in the columns of the BEE JOURNAL the hitherto puzzling problem: "What is royal jelly, that substance known to produce the transformation of worker larvæ to queens?" Profound scientists of Europe and this country have delved into the secrets of the grand problem, but none of them have handed down a satisfactory solution. Yet, it does not seem rational that the question is so obtruse as to forever remain past finding out what the so-called royal jelly consists of; the source from which it is derived; its definite action on larvæ; and whether it is administered by the workers as a nourishing aliment to larvæ; in royal cells, or for the purpose of impregnating the larvæ; (as pistilliferous flowers are impregnated with pollen) and thus develop a female bee fully qualified to reproduce males. The settled doctrine of writers on bee-matters is that it is chiefly due to the excess of food served to the larva by the workers that produces the transformation from worker to queen. Still no writer has ventured to assert that such is a demonstrated fact. The late Baron of Berlepsch, the able expounder of the Dzierzon Theory, and the most scientific

and practical apicultural writer and experienced apiarist in all Europe, wrote thus:

“Every hypothesis, however, yet submitted from any quarter, rest chiefly upon the assumption that the development (of fertile workers and queens) has by some means been over-stimulated for a brief period, and as the result affects the sexual organs more especially, the quantity and quality of the food administered has been looked to as the exciting cause.”

If his assumption be admitted then individual female bees are very likely to be reproduced imperfectly developed in all the degrees between a rudimentary fertile worker up to a perfect queen. Furthermore, were it true that development depends on quantity of food or the over-stimulating caused by high feeding, the workers would be able to supply themselves with queens at all times; when on the contrary it is well known that workers cannot always perfect queens when furnished with everything necessary for that purpose except the impregnating principle—semen.



EGGS AND LARVA.

A full knowledge of the reproduction of the honey bee is of great importance, and at the very foundation of the science of bee-culture and of great value to those who intend to breed the superior races of bees, especially the principles of hybridizing so as to prevent their deterioration and improve the breeds. And it is of great moment to the science of entomology to determine whether insects are produced by parthenogenesis, as is believed, or by semen received by the male progenitors. As for myself, I have conclusive evidence that such queens as are reproduced by furnishing a colony of black bees with eggs laid by an Italian queen, is in some degree hybridized.

All of the points in the “Dzierzon Theory” have been demonstrated except his theory of the reproduction of bees, particularly drones and queens. It seems that he was sorely puzzled in his profound research to comprehend the laws involved in the strange phenomena—virgin queens

reproducing male bees—and to dispose of the (to him) inexplicable point in his colossal theory, he jumped at a conclusion which was based upon the hypothetical doctrine advanced by Professors Von Seibold, Leuckart, and Dr. Donhoff, the fathers of the theory called “Parthenogenesis,” that is procreating without male sperm. It was during the period that Dr. Dzierzon was making public his theory that Mr. Elihu Kirby, of Henrietta, N. Y., attempted to make known the result of his long-time and attentive research into the principles of reproduction of the different races of honey bees. He was a scientific apiarist of long experience, and enthusiastic in the cause of progressive bee-culture. Not until 1861 was there published or circulated in this country a periodical devoted to bee affairs and scarcely no attention was given to scientific bee-culture at that time. Mr. K. communicated to the AMERICAN BEE JOURNAL at different times just after its advent, the discoveries he had made relative to the reproduction of bees, but not much attention was given it further than a brief notice by the editor, the lamented Samuel Wagner, who, like the great Dzierzon, seemed not to comprehend the evolution of the reproduction of insects.

During the period of 1859–63, Mr. Kirby was in failing health, and when in the summer of 1863, he was about to bid adieu to his long-cherished theme and go from the altar of home on earth to a heavenly inheritance, he besought me to further his designs and he committed to my charge his new theory of the reproduction of drones and female bees. The result of the case thus consigned to me is as follows, conclusions that I have come to derive from careful observations for many seasons, viz., *videlicet*.

To produce drones the workers fecundate the worker larvæ in royal cells with drone’s semen, which gives the elements of queens. The workers supply the said larva with animal secretion, water, bee-bread and honey, until it secretes

sufficient material for a queen, and when the larva arrives at maturity it is then metamorphosed to an egg substance, from thence it passes to a chrysalis state, and in the pupa state her ovary is formed and impregnated with semen retained in the larva state imparting the elements of life. She then leaves her cell and is prepared to lay eggs that produce drones only, without further fecundation, and when the drones are matured from their natural genital propensities deposit their semen in the queen's spermatheca to enable her to fecundate her full grown eggs to produce workers, and also deposit semen where the workers can obtain it in the absence of the drones, to perfect queens, and for storing it in their combs, where it retains its vitality at least from the time that the drones are expelled until they are reproduced the following season. It is ascertained that the drones and queens can be hybridized by their drone progenitors in the embryo state, which is conclusive evidence of their being fecundated with drones' semen.

To produce workers the drones deposit their sperm in the queen's spermatheca while on the wing (and on top, clasping the drone's back to herself) and from thence she fecundates full grown eggs, as they pass the mouth of her spermatheca on the way out of her oviduct, and by the combining of the elements of the drone and worker in one, by which the worker is produced. Thus, there can be no logical reasoning in saying that the workers are produced by semen, and the drones and queens are produced without semen.

To produce queens the worker fecundates the worker larvæ in royal cells with drone's semen which gives the elements of the drone, worker and queen, combined in one, in the larval state; it secretes in its growth the proper material for perfect queens, and when the larva arrives at maturity it is transformed to an egg-form, and then to a chrysalis, and in that state her embryo ovary is formed and impregnates in the upper points or sacks of her ovary, and contains the elements

of myriads of drone egg germs before leaving her cell, and her physiology is changed in her transition from the chrysalis state to a perfect queen, and is qualified before leaving her cell to lay eggs that will produce drones only. To be fully qualified to produce workers she must receive a deposit of semen from the drone in her spermatheca. If once filled with semen it is efficacious through life, and qualifies her to fecundate the full grown drone eggs as they pass the mouth of her spermatheca, and causes them to produce workers, and to lay all the eggs, both male and female and workers, that the colony may require. It is ascertained that the embryo drone, workers and queen can each be hybridized in the ovary, egg or larva state, which is communicated to the whole production. I think the evidence conclusive in the reproduction of the queen. The fertile workers are produced by the workers taking the drone's semen into their stomachs, and from thence it is transmitted to their embryo ovary, and fecundates it, which gives the elements of life to the progeny, and qualifies them to lay eggs which produce drones only, unless the eggs are further fecundated by being brought into contact with semen. It appears that the young queen's ovary on leaving her cell, and the ovary of the fertile worker when fecundated, are identical in the production of drone eggs. Therefore, the evidence is that semen is the agent in both cases.

I wish to call attention particularly to the following points: 1st. The embryo ovary of young queens must be fructified before she leaves her cell with drone's semen, which gives the elements of life to her drone progeny, and forms the basis for the whole progeny of bees. To produce the 3 sexes of bees there are 3 distinct fecundations. 1st. The embryo ovary of the pupa queen to produce the drones. 2d. The full grown egg to produce the workers. 3d. The worker larva is fecundated by the workers with semen, given off by the drones to produce the queens. And all in the larval state the

secrete sufficient material to perfect in their transition either drones, workers or queens, and they each can be hybridized in the embryo state.

2d. In the reproduction of bees there are 2 distinct egg forms: 1st. The eggs that produce the larva. 2d. The larva when it arrives at maturity is transformed to an egg substance, of which it forms the chrysalis that produces the perfect bees and their sexes.

3d. It requires 3 states of existence to perfect the organism of bees. 1st. The larva. 2d. The chrysalis. 3d. The perfect bee. The queen first deposits her eggs in the proper cells or utricles in which the larva is hatched and supplied by the workers with animal secretion and food until their transition to an egg substance or chrysalis.

I will propose the following question for consideration: What is it that is found in the royal jelly that is possessed of such impregnating powers as to cause the ovaries of the workers to produce drone eggs?

Richford, N. Y., March 14, 1881.



For the American Bee Journal.

Putting Wires into Comb Foundation.

J. G. WHITTEN.

Mr. John F. Cowan, in his article on "The Practical use of Foundation," published in the BEE JOURNAL of March 9, says: "It has been practically demonstrated to my satisfaction that these results can only be obtained by Mr. Given's method of introducing the wires, and if by a happy combination the Dunham foundation could be made and wired by the Given or a similar process, the foundation controversy would be virtually ended."

I would like to say to Mr. Cowan and others who may be interested, that last season I hived about 40 full sized natural colonies, on Dunham foundation, in Quinby frames, prepared in the following manner: The frame is wired by sewing in 2 horizontal wires, spaced off so that there will be 3 equal spaces from the top bar down. I use a triangular top bar and fasten the foundation by pressing it down to the bar with the thumb and then running a stream of melted wax and rosin over it. Then by running a wheel, made of a cent, over the wire I imbed the wire into the foundation; this also forms a groove in the foundation in which I run a stream of melted wax which covers the wire, and when drawn out will be perfect and will neither sag nor break out by extracting. There should be a good half inch of space between the foundation and bottom bar, as it will settle enough to bulge the comb if left full length. To give it a thorough trial I hived 2 heavy natural colonies in one hive with the mercury at 90 in the shade and basswood honey coming in very fast, and when drawn out every comb was perfect. By bending a spoon so that it will pour a small stream and with a little practice, you will find it a short task to fasten the foundation in the frames.

Genoa, N. Y., March 14, 1881.



For the American Bee Journal.

Importing Bees from Italy.

CHAS. DADANT.

Mr. A. Salisbury, under the above heading, says that "It is no longer a question: the Italian bee of Italy is not a distinct race.... Later investigation proves the fact that there are black bees in Italy, as anywhere else, even in the vicinity of Rome itself."

Mr. Jones, at the Convention in Cincinnati last fall, asserted that he had seen black bees at several places in Italy, even in the vicinity of Rome. All my inquiries, as well as the reports of prominent and disinterested bee-keepers of Italy, such as Mr. Mona and Dr. Dubini, prove that there are no hybrid bees in Italy, and, of course, no black bees.

Will Mr. Jones tell us in which apiaries he saw black bees? Of course, by black bees we understand entire colonies of black bees. Then, he saw also colonies of hybrid bees, for the mixing could not be prevented. But if Mr. Jones saw only a few black, or *seemingly black* bees, in a colony, this circumstance, caused either by the dark contents of their stomachs, or by some other accidental cause, we cannot infer from it that there are black or impure bees in Italy. I hope that Mr. Jones will answer this question.

Mr. Jones adds that, in his opinion, the Italian bees were descended from the bees of Holy Land, or those on the Island of Cyprus. Such an opinion raises the question: Are the yellow bees from Cyprus, from Syria, or from Italy, the original bees; or the black bees, of more northern climates,

the original bees, the yellow color being only an improvement?

According to the law of natural selection, the yellow bees of these three countries are about similar, because the three countries enjoy a mild climate. The idea of Mr. Jones' that the Italian bees descended from the bees of Cyprus or of Syria, cannot be sustained, for it leads to the idea of large importations of bees from these countries, into Italy, at a time when the means of transportation were few, long and difficult.

The introduction of a few colonies of these bees into Italy would have been unable to effect the smallest change in the race then existing; for by our introduction of Italian bees we have experienced how hard it is to overcome the returning to the type which is prevalent in a country. Besides, although we have had too little time to study the habits of the Cyprian bees, having received our queens last summer only, we have noticed that, while they resemble in color the Italian, their habits are not the same. For instance, the Cyprian bees do not cling to the combs as persistently as do the Italians, and resemble more the blacks in this respect; the Cyprian queens, like the common queens, are more easily frightened, and more difficult to find, than the Italian queens.

As to their other qualities we are unable to say anything. It will take a few seasons to test them thoroughly. It is, therefore, desirable to see them tested by a great number of bee-keepers in comparison with Italian bees.

I read in the Italian bee paper, *L'Apicoltore*, for January, just received, that the Central Society of Italian Bee-Keepers will have an exhibition on the first of May, to which the bee-keepers are invited to send bees from every part of the country (probably to answer the assertion of Mr. Jones, that there are black bees in Italy), in order to compare the varieties which can exist on the entire peninsula. The report

of the commission of this society will thus put an end to the discussions between those who contend that there are black bees in Italy, and those who say that the Italian bees are all pure. Yet, it is well to remember here, that in Italy, as well as in Germany, they count but two yellow rings; for they do not count as a ring the first segment, to which the thorax is attached.

Hamilton, Ill., Feb. 5, 1881.

For the American Bee Journal.

Bees and Grapes.

REV. M. MAHIN, D. D.

I notice that the question whether bees destroy sound grapes is again being discussed. I have been a bee-keeper for 11 years and during most of that time have raised grapes enough for family use, and I have given considerable time and attention to the question under discussion. All my observations go to show that bees do not puncture sound grapes. I have seen them sucking the juice from grapes that had been broken by birds, and have picked off the broken grape, and watched the result. The bees would run about over the bunch hunting for an opening, and finally abandon the search. Last season a great many grapes were destroyed or injured in this part of the country, and I gave the matter special attention. Many of the grapes cracked more or less from the effects of rains following dry weather, and many more were broken more or less by birds. As forage was

scarce the bees worked industriously on these broken grapes until they were all gone. But on all the bunches there were some grapes that were not broken, and these remained on the vines until late in the season. After the juice had been sucked from all the broken skins I saw the bees for many days vainly searching for openings from which they might obtain the supplies they had been accustomed to draw from the broken fruit. These sound grapes remained on the vines, in some cases, for weeks after the bees had ceased to get anything from the broken ones. Now it is plain that the juice of these very ripe grapes would have been quite as acceptable to them as that from the ones they are accused of having punctured and destroyed. And to my mind it is clear that if they had punctured and destroyed as many as they are accused of doing, they would not have become suddenly reformed as the grapes became sweeter and more delicious. I will not affirm that the bees cannot puncture the skin of a grape, but I do affirm that as far as my very careful observation enables me to judge, they do not. And if I am correct in this the injury done to the grapes is very small. The injured grapes would spoil in a few days if the bees were not to touch them.

As far as I have been able to observe wasps, hornets, &c., do little injury to grapes. The mischief results mostly from the cracking of the skin, by a very few days, even, of wet weather after it has been dry for some time. The skin of the grapes becomes so full that a jar from the wind or from the alighting of a bird on the bunch, will cause them to crack, and then, if there is a dearth of honey, they are sure to be sucked dry by the bees, with more or less help from yellow jackets, hornets, and wasps. It is possible that in some cases the skins are cut by wasps, &c., but I think the cases are exceptional.

Huntington, Ind., March 4, 1881.

For the American Bee Journal.

The Use of Separators for Box Honey.

GREINER BROTHERS.

In starting an apiary it is of great importance to adopt a hive that will prove satisfactory to the manager, in all its features, for the present as well as for the future. It is not an easy matter after an apiary has been started and hives and appliances have accumulated, to change the sizes or dimensions of such, if they should not be satisfactory. In the different manifestations of the hive we find that it is necessary to have brood frames and sections interchangeable, in fact, it is still more convenient to have all the different parts of the hives as uniform as mechanical workmanship can produce them, so that frames, honey-boards, division-boards, covers, sections, mats, &c., may be picked up anywhere and adjusted to any hive desired.

The use of separators is another feature of this kind; if once adopted and the bees arranged accordingly, it may cause considerable trouble to remodel a lot of appliances, especially if separators of any perceptible thickness are used.

In the BEE JOURNAL for Feb. 2, Mr. Heddon gives some very good hints on "hive and section making," but we can not endorse all his points, and in this article we refer in particular to his closing sentence.

It seems strange to us that Mr. Heddon pronounces separators "nuisances," whilst other prominent bee-keepers,

and we believe the majority, use them and advocate their use. It must certainly be a query to young beginners, who seek information amongst the contributors of the JOURNAL, to encounter such square contradictions. Our experience is about as follows:

The 2 first years of our experience in bee-keeping found us equipped with open surplus cases, we mean by surplus cases the adjustable half-story, with the proper number of frames containing sections. The seasons were good and the crops abundant, but the shape of a good share of our honey was anything but desirable; it was not uniform in thickness nor even; some being thick on one end and thin on the other, some were missed entirely, whilst the adjoining one bulged out to take up the space; in short, the variations were many.

To glass and crate this honey for market cost us considerable trouble and we concluded to try separators. The 25 cases we had prepared and used the next season at our honey apiary proved to be a success; the honey was “just splendid;” the sections in shape, thickness and weight were as near perfect as could be desired, and we decided at once to produce honey in no other way. However, we were not entirely satisfied; we knew separators were objected to by some bee-keepers on account of a smaller yield. Mr. Heddon says, on page 33 of the JOURNAL, “These separators cost me too great a portion of my surplus crop.”

To satisfy ourselves on this point we used the following season about 100 cases, rigged as the first 25, with separators, which we scattered in our different apiaries side by side with open ones. The result was that we noticed very little difference, if any, in the amount of honey stored, and the editor’s opinion, on page 59, was exactly our experience.

Again, Mr. Heddon claims the first cost and trouble of manipulating to be objectionable. We admit separators are an expense, but they need not be very costly. We use basswood,

costing us less than a cent each, and even at twice that cost, would it not be economy then to expend a comparative small amount if we can thereby produce honey in much more attractive shape? Besides we claim separators lessen the trouble of manipulating instead of increasing it. The reason we use wood is because it is cheaper than metal and we believe better adapted, on account of its being the most natural material for bee-habitations.

Since we introduced separators the percentage of unfinished honey is greatly reduced. At the end of the honey season we formerly found open cases almost filled with comb and honey and not one single finished section among them. This is not so much the case since we use separators; when the flow of honey begins to diminish, we have noticed our bees to be at work in a portion of the sections, whilst the remainder would not be occupied at all; we have also taken off cases at the end of a honey flow, which were entirely empty, except 2 or 3 sections, and these were finished and marketable. To be sure these are extremes, but it shows the benefit of separators.

It might appear from the last part of this article, that we apply surplus cases regardless of the working capacity of our colonies. Circumstances may sometimes compel us to do so, but we aim to give our bees no more surplus room than they can occupy.

Naples, N. Y., March 6, 1881.



For the American Bee Journal.

Texas for Bees and Honey.



DR. J. E. LAY.

I write to answer several communications in regard to the adaptability of our great State to bee-keeping, and as apiculture is engrossing the minds of many of the most energetic, progressive and scientific men of our land, I recognize the difficulty of even venturing an opinion. As our great State is so varied in climate and flora, I will state that my remarks have reference to my own section of perhaps a radius of 100 miles. I have lived in Texas since 1850. I passed my boyhood days on her beautiful prairies, amid her thousands of flowers of every hue, freighting our incomparable sea breeze with more than Arcadian sweetness, silence banished from her woodland slopes by the joyous carol of beautiful song birds. Ever delighting in the marvelous beauties of nature, how could I fail to love so beauteous a sunlit home? Yes, and as a grown up boy I love it still. Greek nor Roman, not even Wm. Tell, loved his country better than I, therefore my bee-keeping friends will pardon me if I seem to color a little too strongly. Our State is being filled with energetic farmers who are reaping rich harvests from the virgin soil, for nearly all kinds of seeds that are sown spring forth under the genial rays of the sun to 60 and an hundred fold.

Reasoning by analogy I opine that bee-keeping will result in like manner. Apiculture is in its nascent form here, but the sun of science begins to warm its quickening form. I have studied the best works on apiculture, but have not given it a thorough practical test yet; I purpose doing so this season. There are but few bees in our country, all blacks except my little apiary of 7 colonies, which consists of hybrids and blacks. I intend to Italianize in March, for they indeed possess many advantages over the blacks.

I have just wintered successfully in simplicity hives (plain) without any sort of protection whatever, and this is the coldest winter I ever saw in Texas. Dispatches state that at this time almost the entire North is covered with snow. While my bees were in a quiver of excitement to-day, Feb. 4, bringing in rich loads of pollen and honey from turnips, mustards, &c., I could but delight in their rush of joy. How different is the climate over which our vast brotherhood reaches! Our honey plants reach nearly through the entire year, yielding as good nectar as ever tickled the palate of man. In fact the harvest for bees is almost endless, better, of course, some months. The market for honey has never been developed: a few old "gums" to "rob" for "big meeting" or for some extraordinary visitor is about all ever obtained. "Bees do no good here these days, the moth destroy them," say the "old settlers." The moth skulks away in the light of scientific bee-keeping and its depredations are *nil*.

To be successful all should study the science, read good books on the subject, learn by close practical observation, read the periodicals of our wide awake bee-men, among which there is none better than the *AMERICAN BEE JOURNAL*. Energy and perseverance alone will succeed even in the "sunlit clime" of Texas. Without these, all will just as surely retrograde.

Hallettsville, Texas.



For the American Bee Journal.

Alsike Clover as a Honey Plant.



L. JAMES.

Much has been written for the JOURNAL about the value of this variety of clover for its yield of honey and hay. Such has not been my experience with it, sown on 19 acres of land, and extending over 11 years. In 1869 I bought some 38 pounds of the seed of Mr. Thomas, of Canada. The cost to me, of the seed, duty and express charges, was \$18. Having 9 acres of ground planted with apple trees that had been bearing for some time, and wishing to seed it down to grass I had the ground well prepared for the reception of the seed, and a good rain fell just after it was brushed in; it came up nicely, and as there was favorable rains all through the summer it grew finely. The following season it grew in length of stem and quantity of bloom far beyond my expectation, and when in its full bloom it was a beautiful sight, resembling an ocean of blossoms, and as I looked upon it, you may rely upon it my calculations of boxes of nice alsike clover honey loomed up in large proportions, but like many another calculation based upon what our bees are going to do, it was all in fancy and I was doomed to disappointment.

Day after day their flight was just in the opposite direction, with only here and there a bee to be seen on it. There was a body of timber $\frac{3}{4}$ of a mile distant in the direction they were flying with pastures well set in white clover between this timber and the apiary, and I supposed the white clover pastures was the source of honey supply. This state of things continued for some time, and seeing a bee-man pass by that lived in the timber I inquired how his bees were getting along. He replied they were doing finely as they ought to, for he had never seen heavier honey dews. That was the secret, and soon my boxes began to show evidence of the dark stuff being put into them, instead of alsike honey.

Fortunately for me, before much of it was stored in the boxes, some heavy dashing rains washed it from the leaves and there was no more of the dew for them to gather. The alsike and white clover were in bloom for some time after this, but for some cause the bees paid but little attention to it, and I was vexed to see the promise of a rich return for my expenditure frustrated. I took it for granted that the season was not congenial for its production of honey, as I knew the same to be the case with white clover, as it was last summer. After this at different times I sowed 2 other orchards of 5 acres each with alsike, neither of which did as well as the first piece sown, want of timely rains, &c., being the cause, but by continuous sowing I succeeded in having them tolerably well set with it.

Receiving no perceptible benefit from it, commensurate with its trouble and expense, I have for some time been satisfied that in central Illinois where our white clover is so abundantly furnished in our pastures and road sides, without any expense, and hardy at that, it is time and money put to a poor use.

As a hay producing plant it amounts to but little after the first season, as it becomes dwarfed in habit, and, I believe, will eventually be but little larger in growth under like circumstances than the white variety. The white clover is the honey plant for our latitude, and I presume the alsike for Sweden, from whence it came, and corresponding latitudes. After having had 11 years' experience with it I think it unworthy of attention from bee-men, either for honey or hay; at least where the hardy white clover comes spontaneously to our hands.

There is one thing I ought not to omit, in sowing this Canadian seed I introduced a kind of cockle (different from any I have seen in Penn. or Ohio) that holds its own much

better than the clover, and I begin to think it will be a standing pest difficult to get rid of.

Atlanta, Ill.



Read before the N. E. Convention.

The Supply and Queen Trade.

A. B. WEED.

This is a subject, I believe, of interest to all who are engaged in apiculture, either as bee-keepers, supply dealers or queen breeders, and is growing as the business extends.

The supply business seems as yet to be in a crude state, and prices lack uniformity. In many cases we find needless “cutting” of prices. It may be said that this is a good thing for buyers; but I believe that the opposite is the case, for the inevitable result of unreasonably low prices is inferior goods. When prices are so lowered that there is no margin left for profit, the trade will not be supported with the enterprise which is necessary to stimulate improvements or inventions, or even to put the business on a good footing. The character of the business can best be maintained if the energy of manufacturers is directed to the perfecting of goods rather than the cheapening of them. Good tools are necessary in any pursuit, and seem to be associated with a thrifty business; in fact, the prosperity of a business is largely dependent upon the means at hand of carrying it on. If one tool is better than another—even if the difference is

slight—it is worth very much more, for the benefit of the difference is felt every time that it is used. A good thing may be a source of profit, and a poor one of loss. The best is *always* the cheapest.

There is one respect in which the business is in a better condition than many others, and that is, that there is but very little credit given. This is an advantage to both parties, for the seller loses nothing through bad debts, and the prompt buyer does not have to pay for the losses caused by the careless or dishonest ones.

It is quite common among supply dealers to guarantee safe arrival of goods. This condition of sale is unnecessary, as the express receipt is sufficient, and in case of injury or loss the fact is more readily proven and damages more easily collected than could be from some dealers. It is unreasonable to expect the dealer to be responsible for goods after they have left his hands, especially when the consignee can adjust any difficulty more easily at his end of the line; this is the customary rule in business. When articles are sent by mail the buyer can protect himself against loss by having the article registered; but the precaution is almost unnecessary, as it is *very rarely* that anything is lost in the mails. Of course the sender is required to use necessary care in packing; with most shippers this is a point of pride.

The traffic in queens seems to be closely allied to the supply business—at least so I have found it—for as the bee-keeper begins to feel the need of good tools he sees the advantage of good stock as well; and he naturally looks in the same direction for both. I believe that I express the opinion of the best queen breeders when I say that it is much more satisfactory to sell a good queen at a correspondingly good price—even if the profit is no greater in proportion—than a cheap and poor one, for the reason that a queen, wherever she goes, will represent the stock from which she came. And

I believe, too, that I speak the opinion of all observing apiarists when I say that it pays infinitely better to keep good queens than poor ones. Thus it is that good queens at good prices are more profitable to both parties. Some of the best apiarists have discontinued selling any queens that are not possessed of a high degree of merit, and send out only those which are thoroughly tested and found to be good. In return they receive a suitable price from appreciative customers. This is notably the case in localities where honey raising is an established business, and the value of good stock is therefore understood. It is now almost universally held by apiarists that if good queens are to be obtained they must be raised under favorable conditions. It is freely admitted that to bring about these conditions requires a large outlay of time and thought, as well as money. This especially is the case when queens are to be reared out of season.

The cost of rearing queens will decide their price, for of course they will not be sold at prices which do not pay for rearing and a reasonable profit besides. If buyers insist on having cheap queens, they will get them, but their value will be found to correspond with their price. The one-price rule, which is applied to queens throughout the country, has the effect of causing many poor ones to be sold at fair prices, which really should be killed. It has the tendency to discourage the rearing of very superior ones, for as a rule, a thing is no better than its price. When they are all sold at a uniform price it is to be expected they will be nearly alike in merit, as there is no special inducement for the breeder to improve his stock. The uniformity of price probably originated in the supposition that all queens are equally good, whereas experience proves the opposite to be true. A queen that lays even a few more eggs daily than another is much more valuable, for the extra number of eggs will be multiplied by the number of days that she is kept. This difference alone, so often repeated, will in time amount to

more than the price of the queen. A poor queen is kept at a corresponding loss, although both may have sold at the same price. There are such things as plus and minus outside of algebra. The buying of queens at present has some resemblance to a lottery. They should be graded—at least so far as this is possible—and priced accordingly.

Combinations for the maintenance of artificial prices are impracticable and undesirable. I would only submit that prices be based upon cost of production and a *reasonable* profit.

Detroit, Mich.



For the American Bee Journal.

Who is to Blame for the Losses?

C. H. DIBBERN.

Already the reports of fearful losses are coming in thick and fast. Every severe winter the story is the same. Now the question arises, are these losses of bees inevitable every cold winter? If so then our business as bee-keepers is still a mere matter of luck.

During the last few years of mild winters the out-door wintering men have had things about their own way in our bee-papers. Now, are these papers not a little to blame for admitting articles to their columns giving bad advice to the inexperienced? Many have advocated the wintering on summer stands without protection or care, and persistently

claim to be masters in bee-keeping. I am perfectly willing to admit that bees can be wintered very nicely on summer stands in a mild winter, also that they are wintered successfully if well packed in chaff in a cold winter; but I claim that the labor of preparing them is more than double that of cellar wintering.

I contend that the only certain way is to prepare a suitable place especially for the bees. If a cellar, have the floor cemented and see that it is dry, dark, and well ventilated. In such a place they will not consume more than half the amount of honey they would if left out “packed” in the most approved style. This being a fact they have no particular occasion for a flight. I know that the out-door men claim that cellar-wintered bees do not breed early and are liable to “spring dwindle.” I hardly know what spring dwindling is. By good spring management I have never failed to have my hives crowded as soon as there is anything for the bees to do. Then what is to be gained by having the queen expend her energies and raising vast broods of bees in February to be ready to die when the blossoms come? But sometimes failure comes even in the best of cellars; but would they have fared any better out of doors? Nine times in 10 the cause can be traced to bees filling their hives from the refuse of cider mills. How to keep them from storing such stuff is one of the great problems to be solved.

It is not to be supposed that any kind of a hole under a house will do to winter bees. I have known bees to be packed away among onions, cabbage, and sour kraut. In the spring they wonder what made their bees die. Perhaps they were fastened by wire cloth so that the light could be let in and the bees could not “get out you know.” That such must fail is apparent.

I do not find fault with those who prefer to pack in chaff and winter out of doors; I cannot see, however, that it is the best

way.

It will be the “survival of the fittest” this winter, sure. The box hive men and careless bee-keepers will go out of the business. It is the golden opportunity for the bee-keeper of the future. Soon the fields will be white with the harvest, but the laborers will be few. The bees will have less competition in the fields and the honey in the market.

Milan, Ill.

[Are the papers reprehensible for giving place to candid and respectful arguments, whether based upon tenable or doubtful theories, intended to advance and simplify a science of such magnitude as the bee-keeping interest? Differences of opinion (and honest ones, too,) exist in almost all leading pursuits, and frequently, although seemingly contradicting each other, lead to successful results: again, as has been frequently demonstrated during the past winter, practices embracing all the most approved theories, have alike proved disastrous. There are so many favorable contingencies to be provided, that theories are powerless to insure success. It is interesting, as well as mystifying, to glance through our correspondence from week to week, and note the different methods of preparing bees for winter, and the disasters attending all the different styles. Nor are the cellars exempt from heavy losses, even where success has been proverbial heretofore: The truth is, the winter has been an exceptional one, and loss or success with a single or a few individuals, will neither establish nor disprove theoretical assertions; nor will it justify the “I told you so” class, because successful, in arrogating to themselves all of human wisdom.— ED.]



☞ The North Western Wisconsin Bee-keepers Association will meet at Germania Hall, LaCrosse, Wis., on Tuesday, May 10, at 10 a.m. All interested in bee-keeping are requested to be present.

L. H. PAMMEL, JR., *Sec.*

☞ The next meeting of the N. W. Illinois and S. W. Wisconsin Bee-Keepers' Association, will be held at H. W. Lee's, 2 miles n.w. of Pecatonica, Winnebago county, Ills., on the 17th of May, 1881.

J. STEWART, *Sec.*

☞ On account of unfavorable weather the convention at Monroe Centre, Ill., met on Feb. 8, and there being but few present, adjourned to the same place on March 29, 1881.

A. RICE, *Pres.*



THE AMERICAN BEE JOURNAL

OLDEST BEE PAPER
IN AMERICA
ESTABLISHED
IN 1861.

THOMAS G. NEWMAN.

EDITOR AND PROPRIETOR.

CHICAGO, ILL., MAR. 23, 1881.

Frank Benton in the Far East.

Mr. Jones sends us the annexed extract from a letter of Mr. Benton's, and the following appreciative compliment to the BEE JOURNAL, for which he will accept our thanks:

Herewith I send you extract from a private letter just received from Frank Benton, dated Pointe de Galle, Ceylon, Asia, Jan. 30, 1881. The AMERICAN BEE JOURNAL has a warm corner in my heart. Right glad am I that you have taken time by the forelock, and issued a weekly. I would not have you go back to a monthly for \$25 a year, and you deserve the congratulations of every bee-keeper; that prosperity may crown your efforts is my wish.

D. A. JONES.

FRIEND JONES:—* * * I shall start back with nothing but full colonies. I have seen two native races of bees here, and the comb of a third; one race is stingless, but worthless; the tiniest little fellows, three-sixteenths of an inch long. Another race is *Apis indica*. The third race I do not believe is valuable, since it is a very small bee—smaller than *Apis indica*. *Apis dorsata* is a wonderful bee, whether it can be domesticated or not. It builds in the open air, on branches, often making combs 6 feet long; and I have good authority for saying that 30 natives have each taken a load of honey from one tree. It was not until I reached Colombo that I could find out anything about *Apis dorsata*. I call it *Apis dorsata*, but do not know positively as that is its name, for no one can tell here, and I have not yet seen the bee, as it was too late when I learned where to find it, to go to that part of the Island and reach this French steamer. Everybody says, though, a large bee, from which large quantities of honey are obtained, exists in the interior of the Island. The natives all know it by the name *Bombera*. I start for Singapore by the French steamer “Yangste,” on January 31st.

FRANK BENTON.

Circulars and Price Lists.

—We have received the following Circulars, Price Lists and Catalogues for 1881:

L. H. Pammel & Bros., LaCrosse, Wis.—Italian Queens and Bees—4 pages.

Champion Bee Hive Co., Newcomerstown, O.—Apiarian Supplies—12 pages.

Thomas J. Ward, St. Mary's Ind.—Fruit Trees and Poultry—16 pages.

T. Greiner, Naples, N. Y.—Vegetable and Flower Seeds—24 pages.

D. D. Palmer, New Boston, Ill.—Sweet Home Raspberry—4 pages.

Henry Alley, Wenham, Mass.—Queens and Apiarian Supplies—4 pages.

Wm. W. Cary & Son, Colerain, Mass.—Queens, Bees and Apiarian Supplies—8 pages.

G. W. Thompson, Stelton, N. J.—Bees, Hives and Apiarian Supplies—4 pages.

S. D. McLean & Son, Culleoka, Tenn.—Italian Bees and Queens—1 page.

A. LaMontague, Montreal, Can.—Italian Queens, Hives and Bee-Keepers' Supplies—3 pages.

Jas. J. H. Gregory, Marblehead, Mass.—Vegetable, Flower and Grain Seeds—60 pages.

T. M. Metcalf & Son, St. Paul, Minn.—Field, Garden and Flower Seeds—28 pages.

Nanz & Neuner, Louisville, Ky.—Plants, Seeds. Bulbs, etc.—80 pages.

Landreth's Rural Register and Almanac for 1881, Philadelphia, Pa.—Garden Seeds—70 pages.

Cole & Brother, Pella, Iowa.—Garden and Flower Seeds—44 pages.

J. T. Lovett, Little Silver, N. J.—Choice Small Fruit—40 pages.

Joseph Harris, Rochester. N. Y.—Field, Garden and Flower Seeds—14 pages.

James M. Thornburn & Co., 15 John Street, N. Y.—Seeds for Garden and Farm—96 pages.

L. B. Case's Botanical Index, Richmond, Ind.—A Quarterly Botanical Magazine—40 pages.

☞ The Emperor of Russia, while returning from a review on Sunday, March 12, was killed by a bomb thrown by a Nihilist. He was taken to the Palace and died in a few hours. The assassins have been arrested. His son succeeds him as Alexander III.

An Excellent Suggestion.

Prof. Cook has forwarded us for publication the annexed open letter, addressed to Dr. N. P. Allen, President of the North American Bee-Keepers' Society. The reasons adduced in support of the suggestion are well founded, and must strike all minds favorably. September and October are usually among the busiest months of the year to bee-keepers and farmers, who have their later crops to garner, their honey to take off and prepare for market, their fruit to gather and assort, and their live stock to be made comfortable for winter; while the date proposed by the Professor occurs just at that

period when everybody can spare the time best, when traveling is the most enjoyable, and is quite late enough to enable an approximate estimate of what the harvest will be. It is competent for the Executive Committee (of which President Allen is chairman) to fix upon such time as will best subserve the interests of the Society. We trust they will give the matter an early and careful consideration. Following is the letter:

To Dr. N. P. Allen:

DEAR SIR: As the proposition which I am about to offer is of general interest to the bee-keepers of our country, I beg leave to present it through the AMERICAN BEE JOURNAL:

The American Association for the Advancement of Science convenes at Cincinnati, Ohio, on Wednesday, Aug. 17, 1881. This Association had at its last meeting, in Boston, August, 1880, more than 1,000 members present. Owing to its influence, and the large annual attendance, the local committee at the place where the meetings are to be held are able to procure greatly reduced rates on railroads leading to the place.

Now, I would suggest that the North American Bee-Keepers' Association, which is to be held so near Cincinnati, convene at Lexington on Wednesday and Thursday, August 24th and 25th.

1st. This would accommodate such persons as myself, who wish to attend both meetings, and could not afford time or means were they widely separated by time.

2d. A committee consisting of yourself, Mr. Muth, of Cincinnati, and Mr. Wm. Williamson, of Lexington (I would do what I could to aid), could act in conjunction with the local committee of the A. A. A. of S., and I believe could get the commutation railroad rates to extend to the National Bee-Keepers' Association.

3d. August is a quiet time with bee-keepers, and so far as I can see, nothing would be lost in making the date of our meeting earlier than the usual time.

4th. The fact of accommodating such as wish to attend both meetings, and the reduced railroad rates, could we secure them, would greatly increase the attendance at the Bee-Keepers' Association, and would richly compensate for some loss, if such there would be.

I only make this suggestion, hoping that you and others interested will give it such consideration as its merits deserve.

A. J. COOK,

Vice Pres't of Nat. Association and
President of Michigan Association.

☞ At the Utica Convention, last month, Mr. L. C. Root was appointed a committee to endeavor to have the bill for the prevention of the adulteration of sugar, syrups, etc., then before the Legislature of New York, so amended as to include honey. We learn, with much pleasure, that Mr. Root has succeeded in having it include honey, and Mr. R. is quite sanguine that the bill so amended will become a law of the Empire State. If passed, we hope that the bee-keepers of New York will see to it that it will not be allowed to become a "dead letter" in the statute books of that State.



AMONG OUR EXCHANGES.

GLEANINGS.

Bees and Grapes. —The Klassen and Krock difficulty about the bees of the former committing depredations on the grapes of the latter, is to be submitted to arbitration. It seems that the real trouble was a “personal feud that does not concern bee-keepers at all”—the grape matter was an outgrowth. This matter was referred to in Prof. Cook’s article on page 74 of the BEE JOURNAL, and should now be entirely divorced from the Bee and Grape controversy.

Bees Dead in box hives.—Mr. G. Castello, Saginaw, Mich.,

says that on Feb. 22 he went to a neighbor’s, 5 miles distant, who had a box-hive apiary consisting of 103 colonies of bees. After looking them over, they found only 10 colonies alive; all the rest had died of dysentery.

Honey for sore Eyes. —Mr. S. C. Perry, Portland, Mich., says:

“A neighbor of mine had inflammation in his eyes. He tried many things of many physicians; ‘was nothing better, but rather grew worse,’ until he was almost entirely blind. His family was sick, and I presented him with a pail of honey. What they did not eat he put in his eyes, a drop or two in each eye, 2 or 3 times a day. In 3 months’ time he was able to read coarse print, and now, after 4 months’ use, his eyes

are almost as good as ever. I have also found honey good for common cold-sore eyes.”

MISCELLANEOUS.

Feeding in Winter.—Mr. A. B. Weed, in the *Michigan Farmer*, says:

“Many colonies which were put up for winter with but a small amount of provision, have consumed what was given them, and starved for want of more. Others have but a small amount of stores left, and must be fed soon if they are to be saved. The best way to feed such is to give them frames of well ripened honey, but this the weather will not always permit. The next best thing for them is candy; this can be given at any time, and can be laid on top of the frames. If the cluster is low down in the hive, it should be put down into it, where it can be reached.”

Bees and Grapes. —Mr. W. H. Stout, in the Lancaster, Pa., *Farmer*, gives the following as his experience:

By close investigation I have satisfied myself that bees do not destroy sound grapes. I had, during the past season, 22 colonies of Italian and common black bees; all the hives were in close proximity to the grapes, while a number had the vines trained over them for shade during the heat of summer. The grapes are of the Concord variety, of which I had an abundance of fine fruit, some clusters of which grew within 18 inches of the entrance to the hives. Bunches of the grapes remained on the vines until the frost had killed the foliage, which fell off and left the grapes exposed, affording every temptation to the bees; and this, too, through a season when the honey yield from natural sources was so small that

the bees consumed stores they had gathered earlier in the season. But the bees do work on grapes, and also on other fruits under certain conditions. If the skin of grapes, peaches, pears, etc., is ruptured from any cause, the bees, wasps, ants, etc., are very quick in discovering it, and soon leave only the dried shells. During the hot weather of August, especially when there are frequent showers, the skin of ripening fruit cracks, for reasons which I will leave to some philosophical friend to explain. My conclusions are not hasty; nor were my observations superficial; but they were prolonged from the time the first grapes ripened until the close of the season. I found some clusters of grapes literally covered with bees scrambling and fighting for the little sweets contained in the cracked grapes, which are the only ones on which they work, as I found out by driving the bees away and removing from the clusters all the bursted grapes, when the bees, as soon as they found only sound fruit remained, went away and left the grapes uninjured. We also laid some bunches of grapes on top of the hives and others close to the entrances, also left clusters hanging on the vines close to the hives, where they remained uninjured by the bees as long as the fruit was sound. I know very well that bees can gnaw through heavy muslin, or shave off wood and straw. To cover the bees we have quilts made of heavy muslin, which they sometimes bite through, and we have wood and straw hives on which they have enlarged the entrances; but, nevertheless, I am fully satisfied they do no injury whatever to sound fruit.

Feeding Rye-Meal. —In the *Indiana Farmer* Mr. F. L. Dougherty says:

“Bees will not raise brood without pollen in some shape. We frequently find colonies with but very little, and at times

none at all. In crowding them on a few frames, quite frequently those left in the hive contain but little, if any. So it becomes necessary to furnish it to them, until they can gather it from natural sources. Unbolted rye-meal is probably the best substitute, although they will use wheat-flour, corn-meal, oat-meal, or in lieu of any of these, will even carry saw-dust. To get the bees started, place a piece of comb on the meal, and if the weather be pleasant and no pollen to be had they will soon appropriate it. They will leave the meal when natural pollen makes its appearance.”

☞ That excellent Monthly, published in Nyon, Switzerland, by Mons. E. Bertrand,—the “*Bulletin D’Apiculteur pour la Suisse*”—gives the Weekly BEE JOURNAL the following kind notice:

“We have received the first 2 numbers of the AMERICAN BEE JOURNAL, which has been transferred from a Monthly to a Weekly, by its Editor, Mr. T. G. Newman. Only one apiarian publication is issued every 2 weeks, the *Bienen-Zeitung* of Eickstadt. That of Mr. Newman’s is, therefore, ‘the only one in the entire world which is published weekly.’ It is also, without doubt, the most universal. Its principal contributors are among the most distinguished bee-keepers of America, together with scientists, entomologists, chemists and farmers; and the number of those who send it communications can be called legion. It is, with an understanding of the full extent of the services which it renders, through the abundance of the observations and of the information which it brings before its readers, that we offer to our colleague and friend our warmest felicitations on the occasion of the new development of his publication.”

This very kind notice is the more valuable as Mons. E. Bertrand is a man of intelligence and wealth, whose sole interest is his love of

the pursuit of bee-keeping.

☞ *L'Apicoltore*, the organ of the Central Società d'Apicoltore d'Italia, also gives the BEE JOURNAL the following very kind notice, in its excellent number for February:

“The bee-papers are every day augmenting to suit the increasing need of the readers, and the publisher of the AMERICAN BEE JOURNAL, Signor Newman, who came to Europe and to Milan last year, announces that at the beginning of 1881 his Monthly JOURNAL will be issued every week.”

☞ In Mr. A. Hoke's letter, on page 77, he stated that the dead bees covered the ground for several yards. That was bad enough, but our compositor made it a hundred times worse by adding the word *hundred*. The reader will please discount that expression accordingly.



SELECTIONS FROM OUR LETTER BOX

But Few Bees Lost.— We have had a pretty hard winter for bees, although I have heard of but few losses in this section. My bees are packed in chaff, and are all alive but 2 colonies, which were very weak when packed. Success to the BEE JOURNAL.

F. W. BURTNETTE.

Morrice, Mich., March 12, 1881.

An Old Queen.—We have had a couple of warm, bright days at last, and my bees are flying, what of them are alive. Out of 33 colonies, I think I have 10 or 12 alive, some of them pretty strong, others weak. I have 3 Italian colonies—they seem strongest. What hives I have looked into, where the bees are dead, appear to have plenty of honey, and the other bees appear to be taking the honey out, and I fear are taking from the weak colonies also. Should I prevent them from appropriating it? I noticed some drones with one of my Italian colonies; what does that mean at this time of year? I have been a short distance south, returning home 3 weeks ago. There has been great loss of bees in Fayette and Wayne counties, as well as in Wabash. Please answer above questions in the BEE JOURNAL.

JOEL BREWER.

Lincolnville, Ind., March 10, 1881.

[It is not advisable to let bees have access to combs in other hives; if they need honey, put the combs in the hives where wanted, and not too many. If the strong are robbing the weaker colonies, exchange stands with them. The presence of drones thus early indicates an old or defective queen. Unless there is a large quantity of sealed worker brood (indicating the queen is perfect), we would supersede her as soon as possible, unless the bees save the trouble. —ED.]

Gathering Pollen.—My bees gathered pollen lively to-day, and are strong for this time of year. My loss in wintering is 4 colonies, leaving 8 to commence the season with. Nearly all the bees in this county are dead.

JOHN C. GILLILAND.

Bloomfield, Ind., March 15, 1881.

No Winter Flight Yet.—I am trying to winter 163 colonies in Mitchell hives. All are boxed and packed in chaff with 2 thicknesses of burlaps over the bees; the ends of the hives to the division-boards are filled with chaff; combs contracted to such numbers as bees would cover. They were put into winter quarters Nov. 13, and have had no flight yet. I find many colonies affected with dysentery, and 12 are dead. It is snowing to-day with prospects of another blizzard. I cannot estimate the loss at present; will report at a future time. With many others, I am free to throw in my mite of joy for the weekly visitations of the JOURNAL.

D. VIDETO.

North East, Pa., March 15, 1881.

Bees Confined 4½ Months.—This has been the severest winter that I can remember. My 27 colonies of bees have not had a flight since Nov. 1. They are in a cellar; one of my neighbors had over 50 colonies, but there are only 5 left. He tried to winter out of doors, but has put what he had left in a cider mill. Another had over 20 colonies, wintered out of doors and lost all. I have but little hopes of having over 6 or 8 colonies; there is but little hopes of having weather that bees can have a flight for 2 weeks yet. We are in a snow blockade yet. We have had but one mail in over 2 weeks. I like the Weekly better each number; it brings us nearer together and we can sympathize with our bee-keeping friends. Let us hope for the best; there are better times coming. Success to the Weekly.

E. BUMP.

Waterloo, Wis., March 14, 1881.

Closed out by Fire.—I had the misfortune to be “closed out” of the bee-business by fire, on the night of March 4, losing all of my 36 colonies of Italians, one of which contained an imported queen. They were all in the cellar; I also lost all the implements necessary to carry on the business, my house and contents. This was “closing out” rather unexpectedly, but I hope not to remain out very long.

WM. H. TRAVIS.

Brandon, Mich., March 10, 1881.

Bees in Good Condition.—Though there is a great loss of bees hereabouts, mine are yet in good condition, and I hope

they will come out right in the spring. The Weekly BEE JOURNAL I value more and more all the time.

THOMAS LASHBROOK.

Waverly, Iowa, March 11, 1881.

Lost 8 out of 37 in Wintering.—I put 37 colonies into winter quarters, all in good condition except 4 or 5 small late swarms, and as it was a poor season for honey, they did not fill up; 29 were packed under a shed, open to the south and east. Before packing I removed the outside frames and put in cushions made by covering empty frames with sacking and filling with chaff; also 2 inches of the same on the top of the racks. My loss to date is 3. I prepared 5 in the same manner, but left them on the summer stands; lost 4. Two that I was sure would starve if not fed, I removed to a room over another where a fire is kept, placed them at a window and arranged a passage leading outside; then, with wire cloth over the frames, I can feed and examine without their flying out. They are all right. One I left on the summer stand with a set of section boxes, unprotected, and it is very strong. On March 9th my bees had their first good flight since Oct. 25. I had one colony in a box-hive; of course they are dead. Total loss to date, 8 out of 37. Nearly all are strong now. I am with the majority when I say that the Weekly BEE JOURNAL is a decided improvement. Success to it.

WM. MORHOUS.

Dearborn, Mich., March 14, 1881.

Sweet Clover.—Must the sweet clover be sowed over again, or does it sow itself? Please answer in the Weekly BEE JOURNAL, which I could not do without. It is the best bee paper that is published.

LEWIS SIEGMAN.

Newstadt, Ont., March 11, 1881.

[A good “stand” of sweet clover will sow itself, as there are generally some seeds that do not catch the soil the first season, but germinate the second. It is more satisfactory, however, to plant the second season about half the complement put in the first, after which it will bloom annually, and sow itself.—ED.]

Had a Flight in January.—In the winter of 1879 I put 30 colonies into my cellar; but it was so warm that they were uneasy and I put them back on the summer stands. I lost 10 colonies; I now have 20 colonies, facing the south, sheltered by a board fence on the north and covered with about 18 inches of straw. About 10 days ago they had a nice flight, and I covered them up again. I think of building a house for them facing the south, and boarding up the other 3 sides; I will then cover the hives with about 2 feet of straw, which I can remove on a bright day and give them a flight. I intend to leave the straw on them until warm weather, and thus aid them to keep warm for brood rearing, &c. I wish the BEE JOURNAL success.

T. RICE.

Lenox, Ill., Feb. 4, 1881.

Nearly All Dead.—Bees are nearly all dead in this region. I had 33 colonies last fall and now have but 10; a neighbor had 40 and now has none; another had 44 and now has 2; another had 75, and 3 weeks ago they were reduced to 20. Several have lost all but 1 or 2, and some have lost all.

WM. S. BUCHANAN.

Hartford, Ind., March 14, 1881.

Bokhara Clover.—Please answer the following questions in the JOURNAL:

1. When is the best time to sow Bokhara clover?
2. Should it be sown alone or with a grain crop, or with other kinds of clover?
3. Should it be cut for hay, pastured, or kept for bees only?
4. Which is the best kind of hive for comb honey—a one-story with racks to hold sections, or a two-story, with section boxes put in cases in the upper story?

JOHN H. HEARD.

Flesherton, Ont.

- [1. Early in spring is as good a time as any for planting Bokhara, melilot or sweet clover—we fail to discover any difference in them.
2. For bees alone, sow it alone.
3. If desired for cattle or sheep, sow it with timothy, letting them graze it, as it blooms but little the first season; afterward keep them off.
4. One-story with rack is more easily manipulated.—ED.]
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An Enthusiast.—My apiary is located on a hill-side sloping to the west, and hives fronting south. The Macoupin creek is $\frac{1}{2}$ mile south of it, and several sloughs within a mile, with plenty of soft and hard maple, willows and cotton-wood. I packed rags around and on top of my 13 hives, on their summer stands, on the 25th of October. The bees were in good condition. Only one colony gave any surplus; from that

I took 40 lbs., and left them 35. I examine my bees every week and clean out the dead ones. They had a good flight on the 13th of December, and again on Feb. 22d, when every colony had brood in all stages, and No. 2 was crowded full of young bees, and had a queen cell just ready to put the egg in, which I took off. Feb. 26th was a warm day, and No. 2 sent out a swarm; it was queenless, however, so I sprinkled them with peppermint water and united them with No. 12, which was weak. I do not keep bees for profit in dollars and cents, but for pleasure, as I do love them. I am a merchant, and own 275 acres of land, but being an invalid, look to my bees for recreation. In a radius of 4 miles from my apiary, on Nov. 1st, there were 13 bee-owners, with a total of 73 colonies. On the 1st inst. there were 19 colonies left, and they were in bad condition. I am the only one taking the BEE JOURNAL here—success to it.

R. M. OSBORN.

Kane, Ill., March 4, 1881.

Bees All Dead.—I now send you my report for the winter of 1880–81, which will long be remembered by the bee-keepers in this locality. I commenced the winter with 9 colonies of bees, all carefully packed in chaff on the summer stands with plenty of nice sealed honey. They were packed on the 13th day of last Nov., and from that until the present time (121 days) there has not been a single day that the bees could safely fly, and the consequence is my bees are all dead, from the effects of their long confinement. They left plenty of honey, but the combs are badly soiled. I am not discouraged, however, and shall try again. A gentleman living not far from here had only 8 colonies left out of 39, 2 weeks ago, and when spring condescends to smile on us again we think it will not need a returning board to count the

bees in this county. I am well pleased with the new Weekly; it is always a welcome visitor.

J. R. KILBURN.

Fisher Station, Mich., March 14, 1881.

Bees Robbing.—Here in Texas we have had a severe winter, but not much snow. The thermometer went down to 20° above zero. Last season was a poor one for honey; we had a cold spell in Nov.; then had warm weather for 2 weeks, and my hybrid bees began to rob. The pure Italians behaved well, neither robbed nor let the others rob them. I used water and kerosene oil, but it was of no use; at last I hit upon a remedy. My hives have the bottom boards projecting in front. I ripped out one-inch square pieces 5 inches long, cut coarse wire cloth 2x6, bent it lengthwise in the middle, tacked on 2 sides of each block, leaving wire about 5 inches to give them air; I drove a nail through each end and nailed it in front of each hive. Every 10 or 15 days when the weather was fine, an hour before night, I let them out to have a fly. We have had fine weather for the last 2 weeks. I let the bees out on Jan 30; they have been busy carrying in pollen from elm since Jan. 31, and have forgotten their stealing propensities. I opened some hives this evening and found plenty of sealed brood, and will have drones flying by Feb. 24.

J. W. ECKMAN.

Richmond, Texas, Feb. 10, 1881.

Chloroform.—About 10 years ago I used chloroform in handling bees, after the following plan: I provided myself with a tin slide about 5 inches long and 2 wide; punched a few holes in it, and stitched on one side of it a pad of 3 or 4

thicknesses of cotton cloth. Then after closing all ventilators and entrances except the lower one, I turned about one teaspoonful of chloroform on the pad and slipped it through the entrance, and immediately closed the hive with a wad of cloth, I then listened carefully until the bees had nearly ceased humming (or about 1 or 2 minutes) and then opened the hive and withdrew the slide. They were cross hybrid Italians.

P. F. WHITCOMB.

Lancaster. Wis., March 5, 1881.

Test for Honey.—Bee-keepers need a good honey test, to expose the “rag syrup,” an admixture of honey and glucose, with which the New York market is flooded. In every grocery, meat market and drug store there, can be found cans of “Walker’s best honey,” labeled “Greenpoint, N. Y.,” but there is not much honey in it. Last fall I went into a drug store there with 4 samples of my best honey. They tested it, and what they used turned it perfectly black. I saw one of Walker’s cans of honey there, and asked them to test that; they did so, but the same drugs had no effect whatever on that. They would not tell me what they used to test it; but I would like to have a good and simple test given in the BEE JOURNAL.

H. RICHEY.

Sing Sing, N. Y.

[Pure green tea, well steeped, is used by many to detect the presence of glucose in honey. If the honey dissolves without changing the color of the tea, it is supposed to be pure. But in these days of “enterprise,” it is frequently a matter of doubt whether the tea is pure; again, if, as is claimed, glucose is sometimes

manufactured without leaving sulphuric acid or other deleterious substances in it, then the tea would hardly expose it when mixed with honey. Alcohol is also used to detect the presence of glucose; but besides being frequently inconvenient to obtain, it requires considerable skill in its use.

Thousands of bee-keepers will unite with us in thanking Prof. Kedzie, of the Michigan Agricultural College, for a simple test to detect adulterations in honey and syrups, and instructions for its application.—ED.]

Three-Fourths of the Bees Dead.—The present severe winter has killed $\frac{3}{4}$ of the bees in this section. Bees have not had a thorough cleansing flight since Nov. 8. One apiary of 61 colonies, well packed in chaff and plenty of good stores, will not go through with over 50 per cent. Mine have been confined in the cellar for 118 days, have wintered well so far, but are becoming uneasy.

M. A. GILL.

Viola, Wis., March 13, 1881.

Mortality of Bees in House and Cellar.—I put 60 colonies of bees in a house and cellar last Nov.; 12 of them are dead and I have taken out one-and-a-half bushels of dead bees. Nearly all have the dysentery. I cannot do without the Weekly. I wish it much success.

MILO MUNGER.

Harvard, Ill., Mar. 14, 1881.

Bees Doing Well.—My bees had a nice flight on the 9th, 10th and 11th of this month and are now doing well. It is cold again to-day.

J. R. WAGGONER.

Grantville, Kan., March 12, 1881.

Dwindling in the Cellar.—I put 53 colonies in the cellar, in good condition, which are all alive but one; but there are a great many dead bees on the bottom of the cellar—more than I ever knew before. I gather them up and carry them away occasionally, to prevent their tainting the air. Will the loss of so many weaken the colonies, and what is the cause of it? My bees have not seen the light this winter, yet they seem all right excepting the loss of so many on the cellar bottom.

WM. F. STANDISH.

Evansville, Wis., March 9, 1881.

[If the colonies were very strong, the loss may not be appreciable. The cause may be attributed to age of the bees when put away, and subsequent long confinement; or the cellar may have been too warm at times, and the bees become uneasy.—ED.]

Contradictory Experience.—The poor bees have suffered dreadfully in this locality, and the circumstances and conditions under which some have perished and others survived the past trying season, are so varied that I am quite at a loss what to think about bee preservation during the winter season. I had 12 colonies last fall; I packed 6 with chaff 6 inches thick around them, and have 1 colony left of the lot. There is honey in the combs, but the bees are all

dead. I put 3 colonies in the cellar; 2 of them are alive, but in a bad condition, the combs being dirty and moldy. I left 3 on the summer stands, and 1 is yet alive. None died for want of honey; there was plenty of food for them in the hives. The 6 were put into the chaff in the latter part of November, and taken out on the 8th of March. The combs look clean and free from mold. About a week before I took them out of the chaff I had taken off the front boards, and finding the bees alive, shut them up again. Upon taking them out this was the only colony that was alive. When I took the chaff off, the bees were crowded around the entrance ready to fly, which they did at once, and had a lively time until they were driven inside by the approach of night. Do you think the other 5 colonies were dead the first time I looked at them? They had a passage through the chaff 1 inch high by 4 wide. A friend of mine here had 4 colonies wintered outside, with an old piece of sail-cloth over them, and only lost one, while old bee-keepers, with between 50 and 100 colonies, have lost one half, and others have lost all.

F. A. HUTT.

South Bend, Ont., March 11, 1881.

[Your question is a stunner; we have no data on which to base an intelligent opinion.—ED.]

Wintered Without Loss.—My 27 colonies came through the winter without the loss of a single one, for which I can thank 4 or 5 colonies of Italians, for without them I should not have had honey enough to have kept them through, even a moderate winter, to say nothing of such a stinger as we have had. I have withheld my opinion in regard to the change in the JOURNAL from a monthly to a weekly till I had tried it a couple of months, and will now say that it would be a great disappointment if you were to go back to a monthly. I

am glad that you have so often devoted your first page in each number to the subject of bee-pasturage, for that is, or should be, our leading study now, till we are on surer ground. The best way to make bee-keeping popular is to make it pay; and it will pay if we can get the pasturage every year. I would rather have a tip-top honey plant than an *Apis dorsata*, if it had a tongue long enough to lick the molasses out of the bottom of a 5 gallon keg. We shall have plenty of white clover this year.

WM. CAMM.

Murrayville, Ill., March 12, 1881.

Bees Uneasy in the Cellar.—This has been a very hard winter for bees in this section of the country. Nearly all the bees are dead that were left on the summer stands. I have 40 colonies in the cellar, all alive but restless. They need a cleansing flight very much. The Weekly BEE JOURNAL pleases me very much.

CHAS. H. DOW.

Freedom, N. Y., March 12, 1881.

Bees Much Better Than Expected.—My bees are much better than I had any reason to expect. I left them on their summer stands, and did not even take the tops off, but I have them all off now. I had about 80 and now have 70 colonies in good shape. I find I must either attend to my bees or quit the business, and have made arrangements with a friend who has about the same quantity, who will take charge. We shall call it the “Gipsy Apiary,” and our motto will be, “if the honey will not come to us we will go to the honey.” Mr. Heddon thinks it won’t pay to move for honey, and he is pretty good authority, but we will try. Keep us posted

through the JOURNAL where is the best place to sell honey. Keep the ball rolling in the suppression of adulterated honey, as well as other adulterations.

I. H. SHIMER.

Hillsboro, Ill., March 14, 1881.

Have Young Bees and Brood.—I put 15 colonies of bees into winter quarters and now have 13 in fair condition; some had young bees 2 weeks ago, and all of them have brood. The last 2 years have been very poor for bees; the last the worst, being followed by such a cold and long winter. About one half of the bees in this locality are dead.

G. M. GIVAN.

Moore's Hill, Ind., March 14, 1881.

Bees in the Cellar 135 Days.—I carried 22 colonies of bees out for a flight on March 8. This is the first suitable day for bees to fly there has been here since they were put in the cellar on the last of Oct. They came through the 4½ months' confinement very well, except 2 or 3 third-rate colonies that had more hive room than they could well keep warm through this cold winter, and now they seem to be somewhat reduced in numbers. The day was rather cold, snow did not soften in the shade but the sun shone brightly, "the winds were asleep," and the bees seemed to enjoy the fray, but left a good number of the slain on untrodden snow. They were returned to the cellar at night and will be supplied with water in their hives, hoping to secure the starting of a good cluster of brood before they are placed on their summer stands, about the 1st of May. I usually keep them in confinement without a flight for 5 or 6 months, with good results, but in 1879 brood rearing ceased about the 1st of Sept.; the hives

were destitute of brood when carried out, April 18, and although the hives filled rapidly with brood, before it began to hatch nearly all the old bees were dead, giving me the most disastrous case of spring dwindling that I have known in an experience of 25 years. I hope to avoid such losses in the future.

A. WEBSTER.

E. Roxbury, Vt., March 10, 1881.

The Best Honey for Winter.—By this time I presume all the readers of the BEE JOURNAL know that the winter has been quite severe—about as destructive to the older people as to bees. Bella Lincoln, the oldest bee-keeper in this section of the country, died this winter; and since then nearly all of his 100 colonies of bees have also died. My 60 colonies are in the cellar with chaff over the frames; some are dead, and the entrances to others are soiled, indicating dysentery. Several which had sealed honey stored in the summer are all right. Some worked on a cider mill, but if they have good sealed honey I do not think it makes so much difference about the kind of winter. I like the Weekly BEE JOURNAL, because it “enthuses” me every time I read it. In any kind of business one needs some enthusiasm, at least once a week.

C. F. SMITH, JR.

Vandalia, Mich., March 12, 1881.

Carrying in Pollen.—My 5 colonies of bees wintered well on summer stands, in double-walled Langstroth hives. They are carrying in dark pollen to-day; I think they get it from the maple.

H. H. LITTELL.

Louisville, Ky., March 5, 1881.

Chaff-Packing of Bees Triumphant.—The winter has been a severe one everywhere. Since the 1st of Nov. until the first days of this month my bees had not had a flight. I live in a very high altitude, about the highest good land in the State. The winter begins early and lasts long. We have an abundance of snow now and it is blustering wildly to-day. I despaired of seeing my bees come out alive; they were covered solidly with snow for 3 months, only the tops of the hives being visible. At last the weather softened and I dug away the snow. The next day or 2 the sun came out warmly and my bees began to fly, and greatly to my happy disappointment they are all alive—all that I had out on the summer stands. One only was dull, which I examined and found enfeebled with dysentery, arising from the feed I gave them in the fall. All others were strong. Just 122 days had intervened between the flights. The sick colony has since died, but the others are in the best condition. This success is a tribute to the chaff-packed hive. Is there another record of 122 days' confinement and yet come out strong?

W. S. BLAISDELL.

Randolph, Vt., March 11, 1881.

Look out for the Robbers.—We have had a very hard winter on bees in this section of the country. Bees that were not properly packed for winter are nearly all dead, while those that were properly packed are nearly all in good condition. We are having good weather now and the bees are flying nicely. Those having weak colonies and hives of combs without bees will have to look out for robbers and

keep their small colonies crowded upon as few combs as they can, keeping the entrance contracted, so that only 1 or 2 bees can enter at one time. Hives in which the bees have died should be closed tightly. The Weekly BEE JOURNAL is a welcome visitor. I could not think of doing without it.

J. A. OSBORNE.

Rantoul, Ill., March 17, 1881.

Two-thirds of the Bees have Died.—Over $\frac{2}{3}$ of all the bees in this part of the State are now dead. I have met with a heavy loss, on account of a cider mill that was within 80 rods of my apiary last fall.

HIRAM ROOP.

Carson City, Mich., March 12, 1881.

Bees in Good Condition.—We put out on the summer stands on the 9th and 10th of March, 150 of our 200 colonies that we had in the cellars in good condition. These were the first days that bees could fly with safety since the first of Nov. We have 50 colonies more in one cellar, but as they seem to be doing well, we shall leave them in until it becomes settled weather. We left 9 colonies on their summer stands but the winter was so long and severe that we could not feed them and 3 of them starved. Now we are busy transferring, that is shaking the bees off the combs, cleaning them off and putting them into clean hives. If we find any not strong enough we double them up. We consider ourselves nearly masters of the wintering question, as our real losses for the last 10 years, we think, would not exceed 6 per cent.; in fact we did not lose a colony in winter or spring, until the number had reached about 100. The BEE JOURNAL is a welcome Weekly visitor.

T. S. BULL & SON.

Valparaiso, Ind., March 15, 1881.

Death Reigns among the Bees.—Having made some inquiry concerning the bees within a radius of about 2 miles, I find some bee-keepers, some who keep bees, and those that let the bees keep themselves. Mr. H. had 3 colonies, all are dead; Mr. L. had 7, one left; Mr. D. left his 11 colonies without protection and now has 11 empty hives for sale; Mr. B. let the winters' blast try his 20 colonies and now has 12 empty hives; Mr. F. packed 37 in chaff and has 11 left; Mr. A. put up 57 in complete order, but with all his precaution all are dead; Mr. B. put into winter quarters 73 colonies of fine Italians, 58 of them are dead. I packed in clover-chaff 101 colonies, and 23 have gone the way of all the earth. My bees were confined in their hives from Oct. 20 until March 6. I packed 24 in Langstroth hives with space the whole width of hive left open, to give plenty of fresh air, yet at the same time warm, with a due amount of packing, and in this lot have not lost one colony, and very few bees; but the end is not yet. To-day I found young bees with brood in all stages.

G. W. NAFTZGER.

South Haven, Mich., March 17, 1881.

No Loss In Wintering.—Nearly all the bees in this vicinity that were left to care for themselves are extinct. I had 14 colonies packed comfortably in chaff before the cold weather commenced, and have not lost any yet. I am highly pleased with the Weekly BEE JOURNAL, and wish it great success.

J. P. MOORE.

Morgan, Ky., March 14, 1881.

Poor Season but Fair Profit.—After selling my surplus colonies, I commenced the season of 1880 with 37 colonies in fair condition; increased by division and natural swarming to 63, and 12 nuclei. I reared 30 Cyprian and Italian queens; had 100 Gallup frames of foundation drawn out, and extracted 400 lbs. of honey. Estimating the increase at \$6 per colony, and deducting the expenses, my income for care and labor is \$250, or about \$6.50 for each colony in the spring. I put 75 colonies, in fair condition (including the 12 nuclei), into winter quarters Dec. 8; some were short of stores, and all had poor honey. On March 1st I found 8 colonies and 4 nuclei dead—4 starved and 8 died from the effect of poor honey and long confinement. More of them are diseased and must have a flight soon or die. With the loss of stock already mentioned, and allowing for more to follow, the credit will be cut down to \$3.50 per colony. The season has been the poorest I ever knew, but even \$3.50 is a fair profit on the investment. White clover gave no honey; basswood lasted only 10 days, but yielded well; had it lasted 2 weeks longer I should have had an average yield of honey for the season. Without this flow of basswood honey, the bees must have been fed, but now they have enough stores to carry them through till spring. As the heavy snows have no doubt preserved the clover, the outlook for honey this summer is good. I hardly need say that I am pleased with the Weekly BEE JOURNAL.

T. E. TURNER.

Sussex, Wis., March 1, 1881.

Planting Buckwheat for a Honey Yield.—In answer to Mr. A. Hodges, on page 78, I will say that buckwheat is a

peculiar plant about yielding honey. I have never known it to fail here in yielding enough honey for the bees' winter stores, and usually very much more; in other localities in the same latitude, it cannot be relied on at all for a honey crop. It seems, however, that it never yields through the entire season in which it can be made to bloom. Quite a large amount of it is cultivated every season in my vicinity, much of it generally coming into full bloom as early as the middle of July, yet I have never known it to yield any honey earlier than the 1st of August, and very rarely before the 10th; but when it commences to yield honey, it does so profusely until the plant itself is ripe, or killed by frost. I would say to Mr. Hodges, or any one else intending the sowing of successive crops of buckwheat, that it is useless to sow any early in the season, to blossom before the 1st of August. I am intending to sow about 20 acres of it this season for my bees. I shall put the first crop of it in the ground about June 25; the rest about July 10. That from the last sowing will remain in bloom until frost comes, even if that is delayed later than ordinary.

O. O. POPPLETON.

Williamstown, Iowa, March 9, 1881.

Local Convention Directory.

1881. *Time and Place of Meeting.*

April 2—S. W. Iowa, at Corning, Iowa.

5—Central Kentucky, at Winchester, Ky.

Wm. Williamson, Sec., Lexington, Ky.

7—Union Association, at Eminence, Ky.

E. Drane, Sec. pro tem., Eminence, Ky.

7—N. W. Ohio, at Delta, Ohio.

13—N. W. Missouri, at St. Joseph, Mo.

D. G. Parker, Pres., St. Joseph, Mo.

May 4—Tuscarawas and Muskingum Valley, at
Cambridge,

Guernsey Co., O.

J. A. Bucklew, Sec., Clarks, O.

5—Central Michigan, at Lansing, Mich.

10—Cortland Union, at Cortland, N. Y.

C. M. Bean, Sec., McGrawville, N. Y.

11—S. W. Wisconsin, at Darlington, Wis.

N. E. France, Sec., Platteville, Wis.

12, 13—Texas Bee-Keepers' Association, at
McKinney,

Collin Co., Texas.

W. R. Howard, Sec., Kingston, Hunt
Co., Tex.

Sept. — —National, at Lexington, Ky.

—Kentucky State, at Louisville, Ky.

Oct. 18—Ky. State, in Exposition B'd'g, Louisville,

Ky.

W. Williamson, Sec., Lexington, Ky.

☞ In order to have this Table complete, Secretaries are requested to forward full particulars of time and place of future meetings.—
ED.



CLUBBING LIST.

We supply the Weekly **American Bee Journal** and any of the following periodicals, for 1881, at the prices quoted in the last column of figures. The first column gives the regular price of both:

	<i>Publishers' Price.</i>	<i>Club.</i>
The Weekly Bee Journal (T. G. Newman)		\$2 00
and Gleanings in Bee-Culture (A. I. Root)	3 00	2 75
Bee-Keepers' Magazine (A. J. King)	3 00	2 60
Bee-Keepers' Exchange (J. H. Nellis)	2 75	2 50
The 4 above-named papers	4 75	3 75
Bee-Keepers' Instructor (W. Thomas)	2 50	2 35
Bee-Keepers' Guide (A. G. Hill)	2 50	2 35

The 6 above-named papers	5 75	5 00
Prof. Cook's Manual (bound in cloth)	3 25	3 00
Bee-Culture (T. G. Newman)	2 40	2 25
For Semi-monthly Bee Journal, \$1.00 less.		
For Monthly Bee Journal, \$1.50 less.		



Honey and Beeswax Market.

BUYERS' QUOTATIONS.

CHICAGO.

HONEY.—The market is plentifully supplied with honey, and sales are slow at weak, easy prices. Quotable at 18@20c. for strictly choice white comb in 1 and 2 lb. boxes; at 14@16c. for fair to good in large packages, and at 10@12c. for common dark-colored and broken lots. Extracted, 8@10c.

BEESWAX.—Choice yellow, 20@23c.; dark, 15@17.

NEW YORK.

HONEY.—Best white comb honey, small neat packages, 14@16c.; fair do., 14@16c.; dark do., 11@12; large boxes sell for about 2c. under above. White extracted, 9@10c.; dark, 7@8c.; southern strained, 80@85c.

BEESWAX.—Prime quality, 20@23c.

CINCINNATI.

HONEY.—The market for extracted clover honey is good, at 8@10c. Comb honey is of slow sale at 16c. for the best.

BEESWAX.—18@22c.

C. F. MUTH.

SAN FRANCISCO.

HONEY.—The “Vigilant” takes 600 cases to Liverpool. There is a slightly improved feeling consequent upon a little more inquiry, but prices show no material appreciation. Discouraging reports are received from the southern part of the State, as to the prospects of the coming crop, but other sections give promise of an abundant yield. With a good supply yet on the market, prices are not apt to be buoyant until the anticipated failure is more fully settled. We quote white comb, 12@13c.; dark to good, 9@11c. Extracted, choice to extra white, 5½@6½c.; dark and candied, 5@5½c.

BEESWAX.—22@22½c., as to color.

STEARNS & SMITH, 423 Front Street.

San Francisco, Cal., March 11, 1881.

SPECIAL NOTICES.

☞ Constitutions and By-Laws for local Associations \$2 per 100. The name of the Association printed in the blanks for 50 cents extra.

☞ “What is the meaning of ‘Dec. 81’ after my name on the direction-label of my paper?” This question has been asked by several, and to save answering each one, let us here say: It means that you have paid for the full year, or until “Dec. 31, 1881.” “June 81” means that the first half of the year is paid for, up to “July 1st.” Any other month, the same.

☞ We will send sample copies to any who feel disposed to make up clubs for 1881. There are persons keeping bees in every neighborhood who would be benefited by reading the JOURNAL, and by using a little of the personal influence possessed by almost every one, a club can be gotten up in every neighborhood in America. Farmers have had large crops, high prices, and a good demand for all the products of the farm, therefore can well afford to add the BEE JOURNAL to their list of papers for 1881.

HUNDREDS OF MEN, WOMEN AND CHILDREN rescued from beds of pain, sickness and almost death and made strong and hearty by

Parker's Ginger Tonic are the best evidences in the world of its sterling worth. You can find these in every community.—POST. See advertisement. 9w4t

☞ When changing a post-office address, mention the *old* address as well as the new one.

☞ We have prepared Ribbon Badges for bee-keepers, on which are printed a large bee in gold. Price 10 cents each, or \$8.00 per hundred.

☞ The Volume of the BEE JOURNAL for 1880, bound in stiff paper covers, will be sent by mail, for \$1.50.

☞ Notices and advertisements intended for the Weekly BEE JOURNAL must reach this office by Friday of the week previous.

☞ Instead of sending silver money in letters, procure 1, 2 or 3 cent stamps. We can use them, and it is safer to send such than silver.

LADIES WHO APPRECIATE ELEGANCE and purity are using Parker's Hair Balsam. It is the best article sold for restoring gray hair to its original color and beauty.

☞ The date following the name on the wrapper label of this paper indicates the time to which you have paid. In making remittances, *always* send by postal order, registered letter, or by draft on Chicago or New York. Drafts on other cities, and local checks, are not taken by the banks in this city except at a discount of 25c., to pay expense of collecting them.

PREMIUMS.—For a club of 2, *weekly* we will give a copy of “Bee-Culture;” for a club of 5, *weekly*, we will give a copy of “Cook’s Manual,” bound in cloth; for a club of 6, we give a copy of the JOURNAL for a year *free*. Do not forget that it will pay to devote a few hours to the BEE JOURNAL.

☞ Sample copies of the Weekly BEE JOURNAL will be sent *free* to any names that may be sent in. Any one intending to get up a club can have sample copies sent to the persons they desire to interview, by sending the names to this office.

☞ Any one desiring to get a copy of the Constitution and By-Laws of the National Society, can do so by sending a stamp to this office to pay postage. If they desire to become members, a fee of \$1.00 should accompany it, and the name will be duly recorded. This notice is given at the request of the Executive Committee.

☞ It would save us much trouble, if all would be particular to give their P.O. address and name, when writing to this office. We have

several letters (some inclosing money) that have no name. Many others having no Post-office, County or State. Also, if you live near one post-office and get your mail at another, be sure to give the address we have on our list.

☞ At the Chicago meeting of the National Society we were requested to get photographs of the leading apiarists, to sell to those who wanted them. We can now supply the following at 25 cents each: Dzierzon, the Baron of Berlepsch, and Langstroth. The likeness of Mr. Langstroth we have copied, is one furnished by his daughter, who says, "it is the only one ever taken when he was in good health and spirits." We are glad to be able to secure one of such a satisfactory nature.

☞ We have filled orders for quite a number of Binders for the Weekly BEE JOURNAL. We put the price low, 30 per cent. less than any one else could afford to sell them, for we get them by the quantity at wholesale and sell them at just enough to cover the cost and postage, the latter being 21 to 23 cents, on each. We do this to induce as many as possible to get them, and preserve their Weekly numbers. They are exceedingly convenient; the JOURNAL being always bound and handy for reference. The directions for binding are sent with each one.



GREGORY'S SEED CATALOGUE.

My Annual Catalogue of Vegetable and Flower Seed for 1881, rich in engravings from photographs of the originals, will be sent **FREE** to all who apply. My old customers need not write for it. I offer one of the largest collections of vegetable seed ever sent out by any Seed House in America, a large portion of which were grown on my six seed farms. *Full directions for cultivation on each package.* All seed warranted to be both fresh and true to name, so far, that should it prove otherwise, *I will refill the order gratis.* The original introducer of the Hubbard Squash, Phinney's Melon, Marblehead Cabbages, Mexican Corn, and scores of other vegetables. I invite the patronage of *all who are anxious to have their seed directly from the grower, fresh, true, and of the very best strain.*

NEW VEGETABLES A SPECIALTY.

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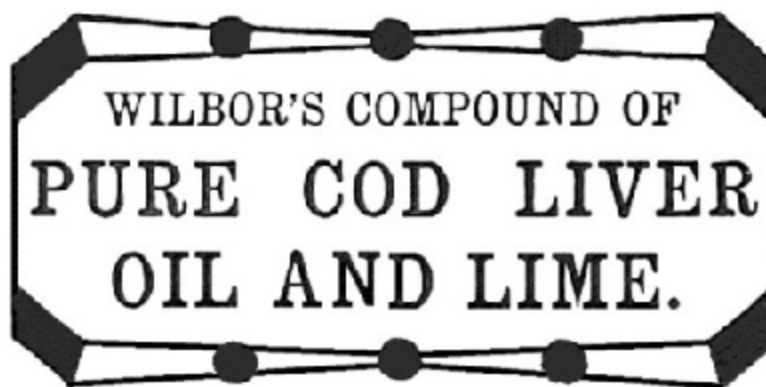
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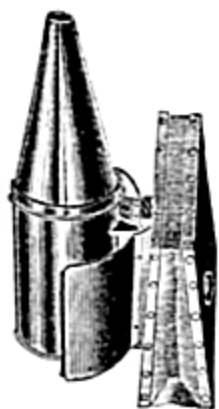
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TRANSCRIBER'S NOTES.

1. **ITALIANS AND HYBRIDS** “—30 or 40 Colonies for sale low.” “low” changed to [“now”](#).
2. Simple spelling, grammar, and typographical errors have been silently corrected.

*** END OF THE PROJECT GUTENBERG EBOOK THE
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